Transformer explainability: a survey

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https://github.com/peulsilva/transformer-explainability

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Introduction and Motivation

- Transformer models excel in NLP and vision tasks, achieving cutting-edge results across various applications.

- Despite their high performance, transformers are hard to interpret due to their deep architectures, large parameter counts, and complex operations.

- Analyzing model behavior is essential not only for performance optimization but also for detecting biases, ensuring fairness, and making AI systems safer and more reliable in real-world applications.

Introduction and Motivation

- **Contributions**:
  - Analyzed Transformer Explainability Methods: Compared attention-based, gradient-based, and perturbation-based metrics.
  - Benchmarked on NLP & Vision Tasks: Conducted experiments on AG News and ImageNet.
  - Probed Attention Heads

Theoretical background

- **Transformers:**
  - Queries, keys and values are obtained from the product of the embedding and trained weights

$$\mathbf{Q} = \mathbf{E}\mathbf{W}^{\mathbf{Q}}$$

$$\mathbf{K} = \mathbf{E}\mathbf{W}^{\mathbf{K}}$$

$$\mathbf{V} = \mathbf{E}\mathbf{W}^{\mathbf{V}}$$

  - The attention matrix is the result of the **scaled dot product** of **Q** and **K**

$$\mathbf{A} = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathbf{T}}}{\sqrt{d_k}}\right)$$

Theoretical background

- **Transformers:**
  - The input passes through layers with residual connections and attention, updating each token embedding with contextual information from neighboring tokens.

$$\mathbf{E}^{(n+1)} = \mathbf{E}^{(n)} + \mathbf{A}^{(n+1)}\mathbf{V}^{(n)}$$

  - Intuitively, we can think of the attention matrix as a matrix corresponding to the importance of tokens across the context

Theoretical background

- **Vision Transformers:**
  - Resize the image to a fixed dimension (e.g., 224 × 224), resulting in an input shape of (batch_size, channels, 224, 224).
  - Split the images into non-overlapping patches (16 x 16).
  - The attention matrix has size (197,197). Why?
  - 196 patches (tokens) of size 16 x 16. Additionally we add the [CLS] token artificially

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Approaches

- Can be divided in 3 sub-domains
  - **Attention based metrics**
    - Studies the attention maps as an indicator of how a token impacts another
  - **Gradient based metrics**
    - Examine the gradient of the loss function (cross-entropy) with respect to a specific token to measure its impact.
  - **Perturbation metrics**
    - Introduce random noise to the embedding of a given token and evaluate its effect on the model's output

Attention based metrics

- **Raw attention**:
  - Uses the raw attention matrix of a given layer as an explainer
  - Limitations:
    - The attention matrix at layer n shows how tokens at that layer affect tokens in the next layer (n+1). However, our goal is to understand how tokens from the initial layer (layer 0) impact the final output.
    - What layer to choose?

Attention based metrics

- **Attention Rollout**:
  - Proposed by *Abnar, S., & Zuidema, W. (2020). Quantifying attention flow in transformers.*
  - Uses the residual connection on transformers to propose a new metric

$$\mathbf{E}^{(n+1)} = \left(\mathbf{I} + \mathbf{A}^{(n+1)}\right) \times \mathbf{E}^{(n)}$$

$$\hat{\mathbf{A}}^{(n)} = \frac{\mathbf{I} + \mathbf{A}^{(n)}}{2}$$

$$\text{Rollout} = \hat{\mathbf{A}}^{(0)} \times \hat{\mathbf{A}}^{(1)} \times \cdots \times \hat{\mathbf{A}}^{(L)}$$

  - Limitations: The equation of $\hat{\mathbf{A}}$ supposes that the residual connection has the same

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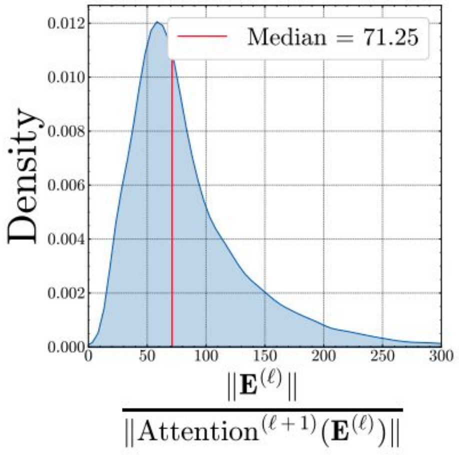
Attention based metrics

- **Attention Rollout**:
  - Limitations: By dividing by 2, the equation of $\hat{\mathbf{A}}$ supposes that the residual connection has the same importance as the attention operation
  - However *Silva et al. (2024)* highlighted that $\mathbf{E}^{(n)}$ has a much higher norm than $\mathbf{A}^{(n+1)}\mathbf{E}^{(n)}$ (see figure below)



Attention based metrics

- **Influence:**
  - Propagates information averaging with the norm of the vectors
  - Initialize the influence with the identity matrix

$$I(\mathbf{E}^{(0)}) = \mathbf{I}$$

$$I(\mathbf{E}^{(n)}) = \frac{I(\mathbf{E}^{(n-1)})||\mathbf{E}^{(n-1)}|| + I(\text{Attention}(\mathbf{E}^{(n-1)}))||\text{Attention}(\mathbf{E}^{(n-1)})||}{||\mathbf{E}^{(n-1)}|| + ||\text{Attention}(\mathbf{E}^{(n-1)})||}$$

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Attention based metrics

- **Influence:**

$$I(\mathbf{E}^{(0)}) = \mathbf{I}$$

$$I(\mathbf{E}^{(n)}) = \frac{I(\mathbf{E}^{(n-1)})||\mathbf{E}^{(n-1)}|| + I(\text{Attention}(\mathbf{E}^{(n-1)}))||\text{Attention}(\mathbf{E}^{(n-1)})||}{||\mathbf{E}^{(n-1)}|| + ||\text{Attention}(\mathbf{E}^{(n-1)})||}$$

$$\text{Attention}(\mathbf{E}) = \mathbf{A} \cdot \mathbf{E}$$

$$I(\text{Attention}(\mathbf{E})) = \frac{\mathbf{A}||\mathbf{E}|| \cdot I(\mathbf{E})}{\|\mathbf{A} \cdot \mathbf{E}\|} \approx \mathbf{A} \cdot I(\mathbf{E})$$

$$\boxed{\begin{aligned} I(\mathbf{E}^{(n)}) &= \frac{I(\mathbf{E}^{(n-1)})r^{(n)} + \mathbf{A}^{(n)} \cdot I(\mathbf{E}^{(n-1)})}{1 + r^{(n)}} \\ r^{(n)} &= \frac{\left\|\mathbf{E}^{(n-1)}\right\|}{\left\|\mathbf{A}^{(n)}\mathbf{E}^{(n-1)}\right\|} \end{aligned}}$$

Gradient based metrics

- **Gradient rollout**
  - Gradient Rollout propagates the gradient of the image with respect to a target class (not necessarily the class of the image)
  - Differently from Attention Rollout, it can be interpreted as “which part of the image is more important for the class c”

Other metrics

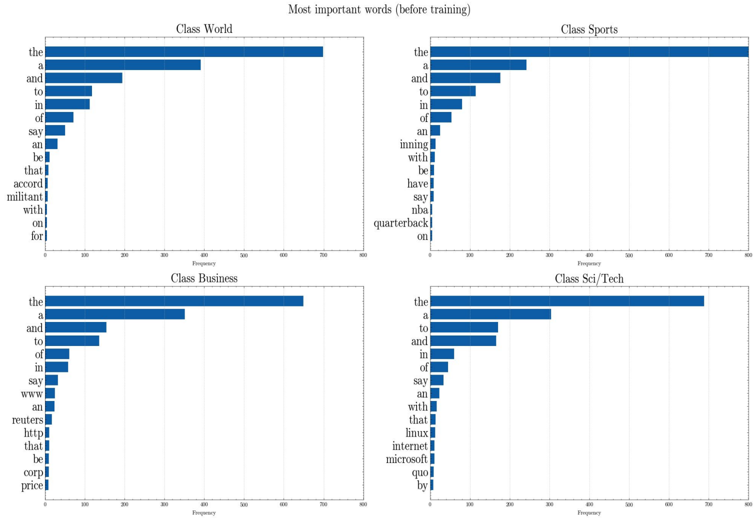
- **Attention Flow**
  - Models the attention mechanism as a directed graph where attention weights act as capacities.
  - Uses a max-flow algorithm to compute information propagation.
  - <u>High computational cost</u>: complexity $O(d^2 n^4)$ versus $O(d n^2)$ for attention rollout and influence.

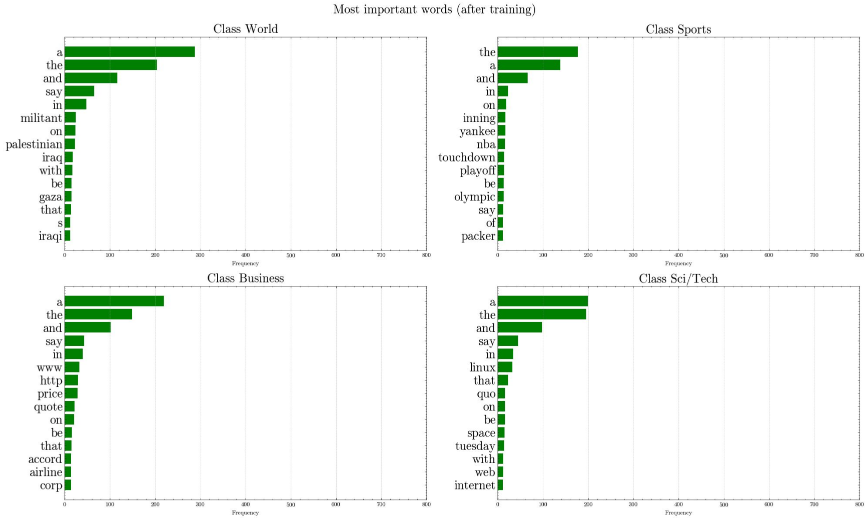
- **Activation Patching**
  - Adds Gaussian noise to specific token embeddings.
  - Measures changes in the model's response due to the perturbation.
  - Provides an estimate of how sensitive the output is to specific token embeddings.

Experiments

- **Qualitative analysis**
  - Study if the metrics makes sense with the output from a human point of view
- **Perturbation metric**
  - Remove the top k most important tokens and assess the impact on performance
- **Extra: Probing attention heads**
  - Identify which attention heads carry the most information

Experiments - Qualitative analysis



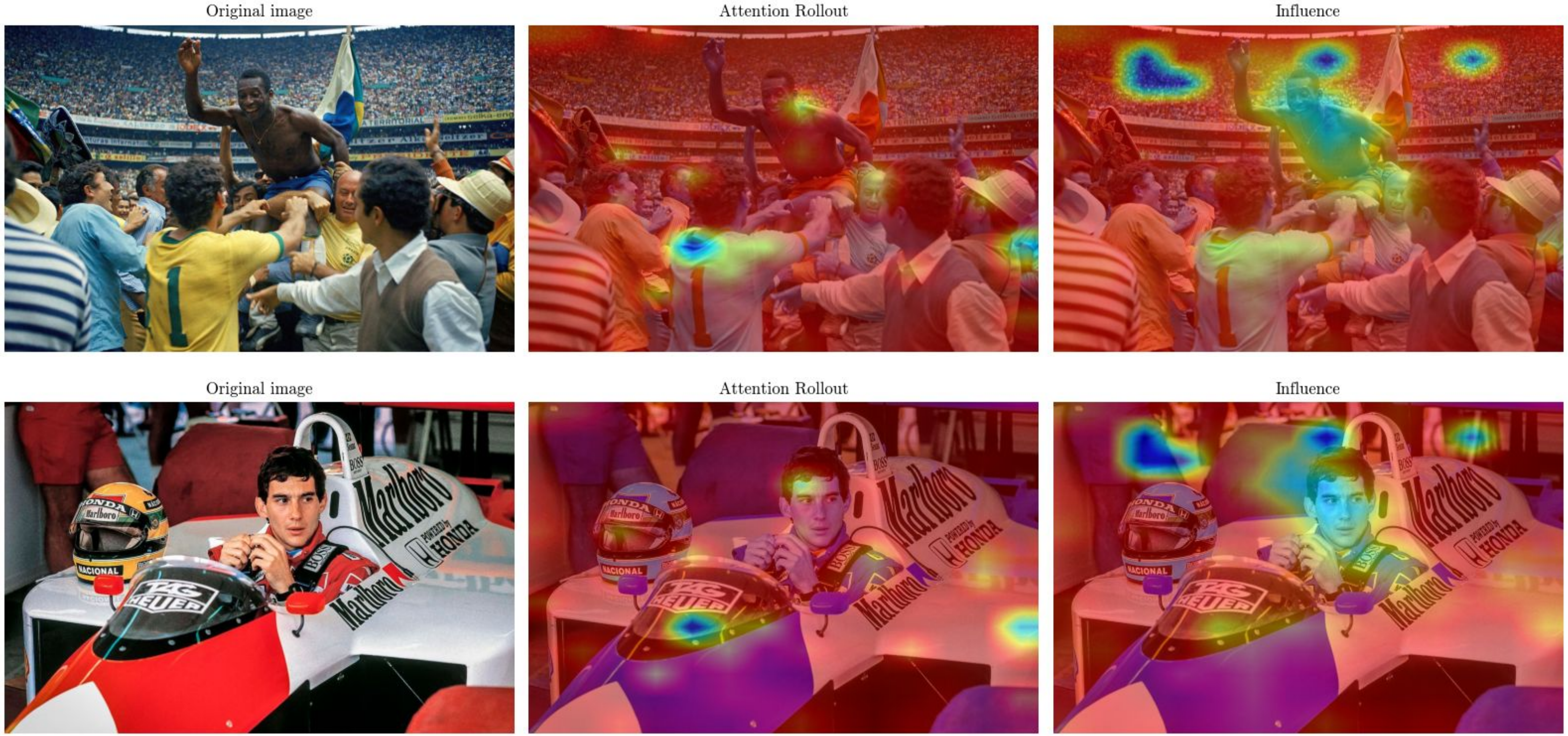


Experiments - Qualitative analysis

Attention rollout vs Influence

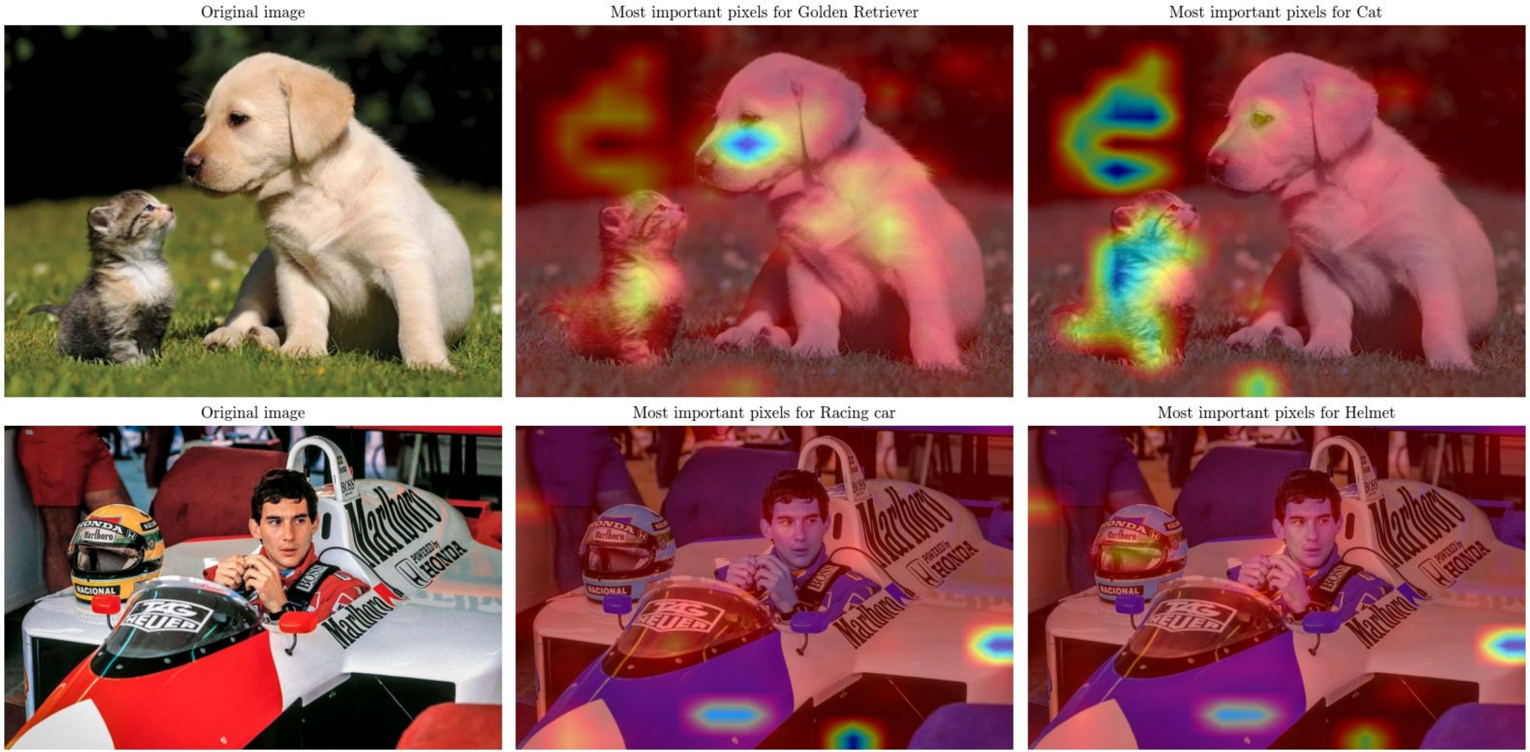
Experiments - Qualitative analysis

Attention rollout vs Influence



Experiments - Qualitative analysis

Gradient rollout for class identification



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Experiments - Perturbation Metric

Setup

- **Positive perturbation**

  For each dataset entry, identify the k% most important tokens relative to the [CLS] token.

  Perform a forward pass through the model without these tokens.

  Expected outcome: A significant and abrupt drop in performance.

- **Negative perturbation**

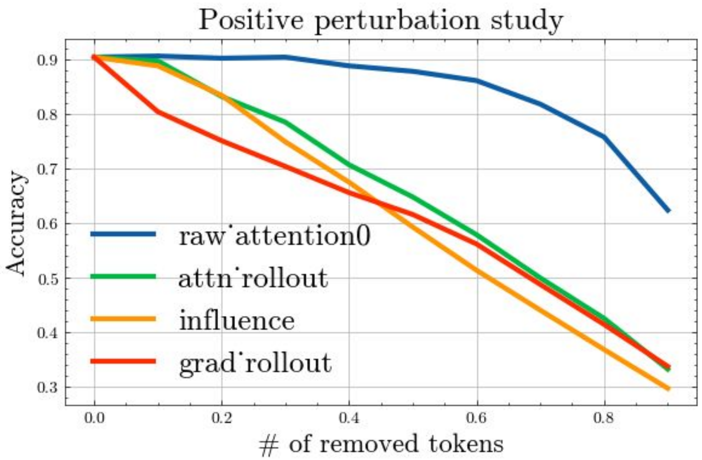
  For each dataset entry, identify the k% less important tokens relative to the [CLS] token.

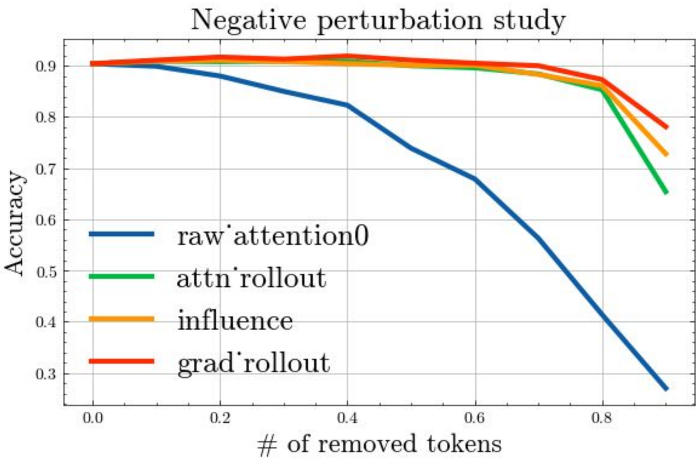
  Perform a forward pass through the model without these tokens.

  Expected outcome: The performance should remain stable until

Experiments - Perturbation Metric

Task: AG News





Metric	AUC (less is better)
Raw attention	0.78
Attention Rollout	0.60
Influence	0.57
Gradient rollout	**0.56**

Metric	AUC (more is better)
Raw attention	0.64
Attention Rollout	0.79
Influence	0.80
Gradient rollout	**0.81**

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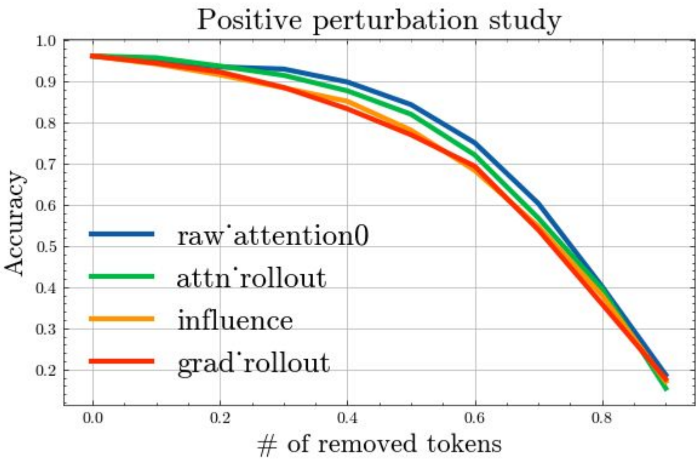
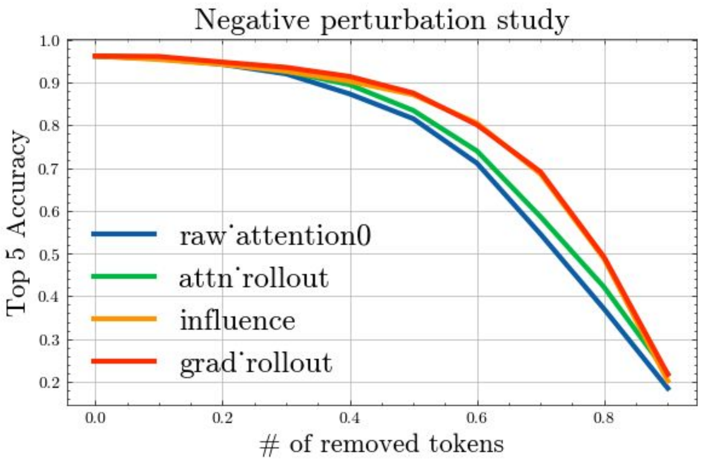
Experiments - Perturbation Metric

Task: ImageNet





Metric	AUC (less is better)
Raw attention	0.69
Attention Rollout	0.68
Influence	0.66
Gradient rollout	**0.65**

Metric	AUC (more is better)
Raw attention	0.67
Attention Rollout	0.68
Influence	0.72
Gradient rollout	**0.72**

Experiments - Perturbation Metric

- As anticipated, Gradient Rollout outperformed in all tests. This is due to the gradients concerning the target class, which correctly identify the most relevant tokens.

- Influence remains a cost-effective metric and still surpasses Attention Rollout. Since it does not require gradient computation, it is well-suited for large-scale models such as Llama and GPT.

Experiments - Probing attention heads

Idea came from *Inference-Time Intervention: Eliciting Truthful Answers from a Language Model* proposed by Li et al. (2024)

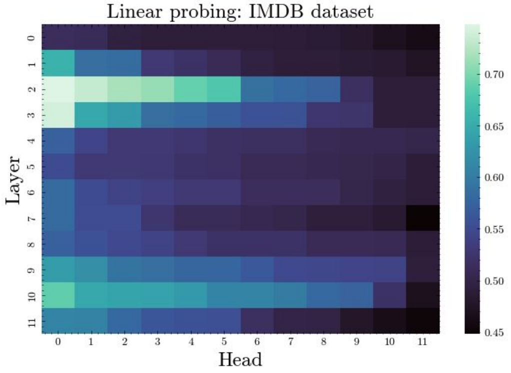
This study aims to identify which attention heads carry the most information relevant to the model's output.

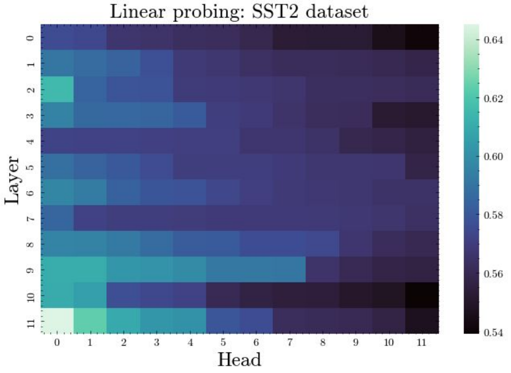
Setup

1. Fine-tuning: We fine-tune a language model on a given text classification dataset.
2. On the validation set, we extract the attention scores of the [CLS] token across all layers and heads.
3. We train linear classifiers for each attention head and evaluate their accuracy on the test set.

Experiments - Probing attention heads

Contrary to our expectations, the last layer does not always hold the most information. Instead, intermediate layers (such as layers 3 and 11 in this case) prove to be the most effective for probing in the two datasets below.





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Conclusion

We studied different explainability methods, including attention-based, gradient-based, and perturbation-based metrics.

Gradient Rollout consistently outperforms other methods in identifying the most important tokens.

Influence remains a computationally efficient alternative that performs better than Attention Rollout, making it suitable for large-scale models like LLaMA and GPT.

Probing Attention Heads showed that intermediate layers often hold more relevant information than the last layer, contrary to common assumptions.